

1)- What is the primary aim of a computer network?

A. To increase the speed of a single computer

B. The sharing of resources among various devices

C. To convert digital data to analog signals

D. To connect only printers and servers

2)- A node in a computer network can be?

A. Only a computer

B. The links connecting devices

C. A computer, printer, or any device capable of sending/receiving data

D. Only a hardware device like a router

3)- What does "NIC" stand for?

A. Network Internet Cable

B. National Interface Card

C. Network Interface Card

D. Network Interaction Component

4)- How does a Hub handle an incoming data request?

A. It sends the request directly to the correct destination.

B. It broadcasts the request to the entire network.

C. It holds the request until the destination is free.

D. It uses a routing table to find the best path.

5)- Why is a Switch considered more advanced than a Hub?

A. A Switch broadcasts to the entire network, while a Hub does not.

B. A Switch delivers the message to the correct destination using a physical address

C. A Switch is obsolete, but a Hub is still commonly used.

D. A Switch connects a LAN to the internet, while a Hub cannot.

6)- What is the main function of a Router?

A. To connect multiple devices within the same LAN only

B. To convert digital signals to analog for telephone lines

C. To connect a LAN to the internet and forward packets to another network

D. To broadcast messages to all devices on a LAN

7)- At which layer of the OSI reference model does a Router work?

A. Layer 1 (Physical Layer)

B. Layer 3 (Network Layer)

C. Layer 2 (Data Link Layer)

D. Layer 7 (Application Layer)

8)- what does a Modem allow a computer to do?

A) Connect to the internet over an existing telephone line

B) Connect multiple computers in a LAN

C) Transmit data using light beams

D) Divide a network connection among multiple devices

9)-What conversion does a Modem perform?

A) It converts the digital data into an analog signal.

B) It converts light beams into digital data.

C) It converts a physical address into an IP address.

D) It converts analog data into a wireless signal.

10)- Which cable type transmits data using light beams and is noted as being the most expensive?

A) Twisted Pair Cable

B) Coaxial Cable

C) Fiber Optic Cable

D) NIC Cable

11)-.....is known as a server while all other computers in the network are called clients.

a)- client

b)-adapter

c)-central controller

12)-there are.....types of networks based on architecture.

a)-1

b)-3

c)-2

13)-..... is a network in which all the computers are linked together with equal privilege and responsibilities for processing the data and has no dedicated server.

a)- Peer-To-Peer network

b)- Client/Server network

c)-Otherwise

14)-..... means that we can add new components to the network.

a)-reliability

b)-scalability

c)-security

15)-By..... We can do business over the internet. For example, amazon.com is doing their business over the internet, i.e., they are doing their business over the internet.

a)- Communication medium

b)- E-commerce

c)- Resource sharing

16)- Which feature of a computer network allows for easy data recovery?

a) Communication speed

b) File sharing

c) Back up and Roll back

d) Scalability

17)-In a Client/Server network, how do clients typically communicate with each other?

a) Directly without any restrictions

b) Through a central server

c) Using external email services

d) They cannot communicate with each other

18)- What is a "Server"?

A) A machine used to access information

B) A central computer that stores information

C) A shared printer

D) An e-commerce website

19)- What does "Scalability" mean?

A) The speed of the network

B) Keeping the network safe

C) Adding new devices to the network

D) Sharing files with others

20)- What is a Peer-To-Peer (P2P) network?

A) A network with one main server

B) A network where all computers are equal

C) A network that is very secure

D) A network that is easy to back up

21)- What is a problem with Peer-To-Peer networks?

A) They are very costly

B) It is hard to back up data

C) They are hard to set up

D) If one computer stops, all computers stop

22)- What is a good thing about Client/Server networks?

A) They are less costly

B) All computers are equal

C) It is easy to back up data

D) There is no main server

23)- Which type of network is used to connect devices within a small area such as a building or office?

- A) PAN
- B) MAN
- C) LAN
- D) WAN

24)- Who was the first research scientist to introduce the idea of a Personal Area Network (PAN)?

- A) Tim Berners-Lee
- B) Thomas Zimmerman
- C) Vint Cerf
- D) Robert Metcalfe

25)- Which of the following is an example of a Wireless Personal Area Network?

- A) Network using USB cables
- B) Network using fiber optic cables
- C) Network using Bluetooth
- D) Network using telephone lines

26)- Which type of network covers a larger geographic area by interconnecting multiple LANs?

- A) PAN
- B) LAN
- C) WAN
- D) MAN

27)- What is one of the main disadvantages of a Wide Area Network (WAN)?

- A) Limited geographical area
- B) Low data transfer speed
- C) High setup cost
- D) Inability to share resources

28)- What does topology define in a network?

- A. The speed of data transmission
- B. The structure of how all components are interconnected
- C. The type of cables used
- D. The number of users in a network

29)- How many types of topologies are there?

- A. One
- B. Two
- C. Three
- D. Four Bus Topology

30)- Which of the following access methods is commonly used in a bus topology?

- A. Token Passing
- B. CSMA (Carrier Sense Multiple Access)
- C. Polling
- D. TDMA (Time Division Multiple Access)

31)- In a bus topology, all the devices are connected through which of the following?

- A. Hub
- B. Switch
- C. Backbone cable
- D. Ring Ring Topology

32)- In a ring topology, data flows in which direction?

- A. Both directions
- B. Random directions
- C. One direction (unidirectional)
- D. Depends on the network type

33)- Which access method is commonly used in ring topology?

- A. CSMA/CD

B. Token passing

C. TDMA

D. Polling

34)-What is a token in a ring topology?

A. A network cable

B. A software program

C. A frame that circulates around the network

D. A hardware device

35)- In a star topology, every node is connected to which of the following?

A. Each other node directly

B. A backbone cable

C. A central hub, switch, or computer

D. A router

36)- The devices connected to the central computer in a star topology are known as

A. Servers

B. Hosts

C. Clients

D. Routers

37)- Which of the following cables are commonly used in star topology?

A. Optical fiber only

B. Coaxial cable or RJ-45 cables

C. Twisted pair only

D. Token ring cables

38)-Which of the following devices is mainly used for connections in a physical star topology?

A. Repeater

B. Bridge

C. Hub or Switch

D. Router

39)- The following statement best describes a Tree Topology:

A) It connects all devices in a circular path

B) It combines the characteristics of Bus and Star topologies

C) It connects devices using a single main cable only

D) It connects devices without any hierarchy

40)- What is the top-most node in a Tree topology called?

A) Root node

B) Parent node

C) Child node

D) Central hub

41)- Which of the following is an advantage of Tree topology?

A) Difficult to detect errors

B) Not easily expandable

C) Easily managed and expandable

D) Completely dependent on one device

42)- In a Mesh topology, how is the total number of connections calculated?

A) $(n + 1)/2$

B) $(n(n-1))/2*$

C) n^2

D) $n/2$

(43) -Which of the following is a main advantage of Mesh topology?

A) High reliability and fast communication

B) Simple and low-cost setup

C) Uses a single central hub

D) Limited fault tolerance

44)- Which of the following is a disadvantage of Mesh topology?

- A) Easy to install
- B) Low cost
- C) Complex and expensive
- D) Slow communication

45)- A Hybrid topology is defined as:

- A) Simple network with one topology type
- B) Combination of two or more different topologies
- C) Wireless network only
- D) Bus network with one main cable

46)- Which of the following is an advantage of Hybrid topology?

- A) Limited scalability
- B) Low reliability
- C) Highly flexible and scalable
- D) Difficult to expand

47)- The following is a disadvantage of Hybrid topology:

- A) Simple design
- B) Expensive hubs and complex structure
- C) Easy to reconfigure
- D) Low installation cost

48)- In Full-Duplex transmission mode, data can be sent:

- A) In one direction only
- B) In both directions but not at the same time
- C) In both directions simultaneously
- D) Only when one device is idle

49)- What does the term "Network Topology" refer to?

- A) The network's transmission speed

- B) The type of cables used in a network
- C) The structure of how network components are interconnected
- D) The type of operating system used

50)- What is the main difference between physical and logical topology?

- A) Physical topology represents routing tables
- B) Logical topology shows device positions, physical shows software layout
- C) Both represent the same network structure
- D) Physical topology shows actual connections, logical shows data flow

51)- In Bus topology, all stations are connected through a single cable called the:

- A) Backbone cable
- B) Fiber trunk
- C) Star connector
- D) Transmission hub

52)- Which device operates at Layer 2 of the OSI model?

- A. Router
- B. Switch
- C. Firewall
- D. Modem

53)- Which of the following classifies networks based on geographic area?

- A. LAN, MAN, WAN
- B. TCP, UDP
- C. HTTP, FTP
- D. SMTP, POP3

54)- A group of devices connected together to share data is called a:

- A. System
- B. Network
- C. Protocol

D. Frame

55)-Which of the following addresses is a valid Network ID?

A. 172.16.100.0

B. 172.16.100.255

C. 172.16.100.1

D. 172.16.100.256

56)-What is the primary purpose of Subnetting?

A. To increase network speed

B. To divide a large network into smaller networks

C. To connect two different networks

D. To increase the number of available IP addresses

57)-How is the Magic Number calculated?

A. 256 - the value of the 3rd octet in the Subnet Mask

B. 255 - the value of the 4th octet in the Subnet Mask

C. 128 - the value of the 2nd octet in the Subnet Mask

D. 64 + the value of the 3rd octet in the Subnet Mask

58)- What is the main goal of Network Management?

A) To install new hardware only

B) To monitor and maintain network performance

C) To increase software size

D) To reduce the number of users

59)- Which of the following is not part of the five FCAPS functions?

A) Fault Management

B) Configuration Management

C) Accounting Management

D) Application Management

60)- In a Network Management System (NMS), what is the function of an "Agent"?

- A) Manages user accounts
- B) Collects and sends data from devices to the manager
- C) Acts as a firewall
- D) Encrypts data packets

61)- SNMP is mainly used for:

- A) Managing hardware installation
- B) Network configuration via XML
- C) Communication between manager and agent
- D) Web server control

62)- NETCONF operates over which secure protocol?

- A) HTTP
- B) SSH
- C) SNMP
- D) FTP

63)- One of the main challenges in modern network management is:

- A) Decreasing automation
- B) Reducing device diversity
- C) Managing cloud and virtual networks
- D) Eliminating monitoring systems

64)- Why are standards important in network management?

- A) To increase company profits
- B) To ensure compatibility and interoperability
- C) To restrict device usage
- D) To improve user interface design

65)- What does ASN.1 stand for?

- A) Abstract Syntax Notation One
- B) Advanced Syntax Network

- C) Application Syntax Node
- D) Abstract System Network

66)-What is the main use of ASN.1 in network management?

- A) Encrypt data
- B) Describe and encode messages in communication protocols
- C) Control routing tables
- D) Monitor user activity

67)- What does MIB stand for?

- A) Management Interface Base
- B) Management Information Base
- C) Message Information Block
- D) Managed Internet Base

68)- What does the MIB store?

- A) User passwords
- B) Network management information (status, performance, faults)
- C) Security keys
- D) Device drivers

69)- What command does SNMP use to get information from a device?

- A) SEND
- B) PING
- C) GET
- D) CONNECT

70)- In the Centralized Network Management Model, what is one disadvantage?

- A) Difficult to configure
- B) Expensive hardware
- C) Single point of failure
- D) Requires many managers

71)- Which model provides better scalability for large networks?

- A) Centralized
- B) Distributed**
- C) Manual
- D) Hybrid

72)- What is a key advantage of Distributed Network Management?

- A) Simplicity
- B) Better scalability and reliability**
- C) Lower performance
- D) No coordination required

73) What is the primary function of Network Management Models?

- A. To define the physical cable specifications and device hardware requirements.
- B. To describe how control and monitoring are organized within a network.**
- C. To standardize the maximum data transfer rates and latency tolerances.
- D. To encrypt all data passing between network devices and servers

74) What is the default subnet mask and the range of the first octet for a Class B IP address?

- A) 255.0.0.0 and 1-126
- B) 255.255.0.0 and 128-191**
- C) 255.255.255.0 and 192-223
- D) 255.255.255.0 and 128-191

75) A network administrator needs to create a subnet for a Class C network that must support at least 60 usable hosts. Which of the following custom subnet masks should be used?

- A) 255.255.255.192**
- B) 255.255.255.224
- C) 255.255.255.240

D) 255.255.255.128

76) Find the number of hosts and the number of subnets in a Class C network with a Subnet Mask of 255.255.255.224.

A. 14 subnets and 14 hosts

B. 30 hosts and 6 subnets

C. 32 hosts and 8 subnets

D. 62 hosts and 2 subnets

77) Find the subnet mask used in a Class B network where the number of devices (hosts) is 1024.

A. 255.255.255.0

B. 255.255.248.0

C. 255.255.252.0

D. 255.255.0.0

78) Find the Network ID IP address: 192.168.1.35/30?

A. 192.168.1.0

B. 192.168.1.34

C. 192.168.1.32

D. 192.168.1.36

79) Find the subnet mask used in a Class B network where the number of devices (hosts) required in each subnet is 459.

A. 255.255.255.0

B. 255.255.252.0

C. 255.255.254.0

D. 255.255.248.0